Three-uniform Berge-C_5 saturation: a two-thirds upper bound, the cone spectrum and exact small values

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Abstract

Let $s(n) = \mathrm{sat}_3(n, \text{Berge-}C_5)$ be the least number of hyperedges of a three-uniform hypergraph on n vertices that contains no Berge five-cycle but contains one after any absent triple is added; neither the hypergraph nor the added triple has to be linear. The explicit upper bound of English, Graber, Kirkpatrick, Methuku and Sullivan has leading coefficient $4/3$. We show that for every $n \geq 47$ every integer m with $\lceil (2n-8)/3 \rceil \leq m \leq n-2$ is the size of a saturated hypergraph on n vertices, so $s(n) \leq \lceil (2n-8)/3 \rceil$; weaker endpoints hold from $n = 7$ on, and the simplest construction glues copies of a seven-vertex block with four hyperedges at one vertex. We classify the saturated hypergraphs in which one vertex lies in every hyperedge: their sizes are exactly the integers from $\lceil 6(n-1)/7 \rceil$ to $n-2$, together with $n-1$ when $4 \mid n-1$, for $n \geq 8$, so such cones cost asymptotically more than the general constructions. We also characterise the saturated loose expansions of bipartite graphs in which one part has all degrees two, and we determine $s(n) = 0, 0, 0, 1, 4, 4, 4, 4, 5, 6, 6, 7$ for $n \leq 11$ from exhaustive certificates. No lower bound beyond small orders is proved; the asymptotic coefficient of $s(n)$ remains open. Lean 4 proves, for actual hypergraphs, Berge cycles and saturation under every absent triple, that every integer from $\lceil 2(n-1)/3 \rceil$ to $n-2$ is realized for $n \geq 14$, that $s(n) \leq \lceil 2(n-1)/3 \rceil$ for $n \geq 7$ and the values $s(n)$ for $n \leq 7$.

1 Definitions and results

All hypergraphs are simple and three-uniform, and the vertex set may contain isolated vertices. A *Berge path of length* ℓ has distinct vertices $v_0, \ldots, v_\ell$ and distinct hyperedges $e_1, \ldots, e_\ell$ with $\{v_{i-1}, v_i\} \subseteq e_i$; we call $v_0, \ldots, v_\ell$ its *cores*. A *Berge-C_5* consists of five distinct vertices $v_0, \ldots, v_4$ and five distinct hyperedges $e_0, \ldots, e_4$ with $\{v_i, v_{i+1}\} \subseteq e_i$, indices modulo 5. A hyperedge may contain further cores. A hypergraph H is *saturated* if it contains no Berge-C_5 (it is *free*) but $H + f$ contains one for every three-element set $f \notin E(H)$. This is the unrestricted notion of [2]: f may share two vertices with a hyperedge of H. We write $s(n)$ for the least number of hyperedges of a saturated hypergraph on n vertices and $\mathrm{Spec}(n)$ for the set of all their sizes.

For uniformity three and cycle length five, English, Graber, Kirkpatrick, Methuku and Sullivan [2, Theorem 5.7(2)] glue copies of the complete hypergraph $K_4^{(3)}$ at a vertex and obtain $s(n) \leq 4\lfloor (n-1)/3 \rfloor + ((n-1) \bmod 3)$ for $n \geq 25$, with leading coefficient $4/3$. English, Gerbner, Methuku and Tait [4] proved that $\mathrm{sat}_k(n, \text{Berge-}F) = O(n)$ for every graph F and $3 \leq k \leq 5$.

Theorem 1. *For every $n \geq 47$,*

$$\left\{ \left\lceil \tfrac{2n-8}{3} \right\rceil, \ldots, n-2 \right\} \subseteq \mathrm{Spec}(n), \qquad \textit{so} \qquad s(n) \leq \left\lceil \tfrac{2n-8}{3} \right\rceil.$$

More precisely, $\mathrm{Spec}(n)$ *contains every integer from $L(n)$ to $n-2$, where $L(n)$ is the smallest of*

the applicable endpoints

$$\begin{array}{ll} \lceil 2(n-1)/3 \rceil & \text{for } n \geq 7, \\ \lceil (2n-4)/3 \rceil & \text{for } n \geq 14, \\ \lceil (2n-6)/3 \rceil & \text{for } n \geq 30 \text{ and } n \in \{25, 28\}, \\ \lceil (2n-8)/3 \rceil & \text{for } n \geq 47 \text{ and } n \in \{41, 44\}. \end{array}$$

Moreover $n-1 \in \mathrm{Spec}(n)$ *whenever* $n \geq 5$ *and* $4 \mid n-1$.

Neither optimality of $L(n)$ nor completeness of the spectrum is claimed. The construction for $\lceil 2(n-1)/3 \rceil$ is a bouquet of copies of the block $\{a,b,c\}, \{a,d,e\}, \{b,d,x\}, \{c,e,y\}$ glued at a (Section 3); the smaller endpoints come from loose expansions of bipartite graphs (Section 4).

A *cone* is a hypergraph with a vertex r in every hyperedge; its *link* is the graph G on $V \setminus \{r\}$ with $xy \in E(G)$ when $\{r,x,y\} \in E(H)$, isolated vertices included. Call a tree *admissible* if it has a centre c with at least three non-leaf neighbours, every neighbour of c is a leaf or has only leaves as further neighbours, and every non-leaf neighbour has at least one leaf; that is, a tree of radius two with at least three branches of length two. It has at least seven vertices.

Theorem 2. *Let* $n \geq 5$ *and let* H *be a cone on* n *vertices with link* G. *Then* H *is saturated if and only if* G *is a nonempty disjoint union of four-cycles and admissible trees, or* G *consists of at least one four-cycle and exactly one further component, which is* K_2 *or an isolated vertex. The sizes of saturated cones are* $\{4\}, \{4\}, \{5\}$ *for* $n = 5, 6, 7$, *and for* $n \geq 8$ *exactly the integers in* $[\lceil 6(n-1)/7 \rceil, n-2]$ *together with* $n-1$ *when* $4 \mid n-1$.

Since $6/7 > 2/3$, a common vertex in all hyperedges is asymptotically costly. The additions in Theorem 2 are arbitrary triples, not only triples through r.

The *loose expansion* of a graph G replaces every edge uv by a hyperedge $\{u, v, \ell_{uv}\}$ with a new *private* vertex ℓ_{uv} for each edge.

Theorem 3. *Let* G *be a bipartite graph with nonempty parts* A, B *in which every vertex of* B *has degree two. Its loose expansion is saturated if and only if* $A = \{a,b,c\}$ *and each of the three classes* B_{ab}, B_{ac}, B_{bc} *of vertices of* B *with the indicated neighbourhoods has at least three elements. Then* $n = 3 + 3(p+q+r)$ *and* $m = 2(p+q+r)$ *with* p, q, r *the class sizes, so* $3m = 2n - 6$, *and the isomorphism classes correspond to the unordered triples* $\{p,q,r\}$ *with* $p, q, r \geq 3$.

Theorem 4. *Let* F *be a nonempty finite* 2*-regular graph on* h *vertices with independence number at most two, and let* $\lambda_{ij} \geq 2$ *for each edge* ij *of* F. *Let* G *be the bipartite graph with parts* $V(F)$ *and* $\{z\} \cup \bigcup_{ij} X_{ij}$, *where* z *is adjacent to every vertex of* F *and each of the* λ_{ij} *vertices of* X_{ij} *is adjacent exactly to* i *and* j. *The loose expansion of* G *is saturated, with* $n = 1 + 2h + 3S$ *and* $m = h + 2S$, $S = \sum \lambda_{ij}$, *so* $3m = 2n - (h+2)$. *The admissible* F *are* C_3, C_4, C_5 *and* $2C_3$.

For $F = C_4, C_5, 2C_3$ Theorem 4 gives $m = (2n-6)/3$, $(2n-7)/3$, $(2n-8)/3$ for all $n \equiv 0, 2, 1 \pmod 3$ from $n = 33, 41, 49$; these are the values $\lceil (2n-8)/3 \rceil$ of Theorem 1. The two-triangle family shows that no inequality $3s(n) \geq 2n - C$ with $C < 8$ holds for all large n.

Theorem 5. $s(n) = 0, 0, 0, 1, 4, 4, 4, 4, 5, 6, 6, 7$ *for* $n = 0, 1, \ldots, 11$. *Up to isomorphism there are exactly* 13 *saturated hypergraphs with four hyperedges, on* $4, 5, 6, 7$ *vertices in numbers* $1, 3, 7, 2$, *and exactly* 32 *with five hyperedges, on* $5, 6, 7, 8$ *vertices in numbers* $1, 2, 14, 15$; *no saturated hypergraph with six hyperedges has* 11 *or more vertices.*

For $7 \leq n \leq 11$ the values equal $\lceil 2(n-1)/3 \rceil$. We do not know the asymptotic coefficient of $s(n)$: the only general lower bound proved here is the trivial $3s(n) \geq n - 1$ of Lemma 7.

2 Basic facts

Call two vertices of H *good* if they are the end vertices of a Berge path of length four in H.

Lemma 6. *Let H be free and $f \notin E(H)$ a triple. Then $H + f$ contains a Berge-C_5 if and only if f contains a good pair of H. Hence a free H is saturated if and only if every absent triple contains a good pair.*

Proof. A Berge-C_5 of $H + f$ uses f, since H is free; deleting f leaves a Berge path of length four between the two cores that f supplies. Conversely a Berge path of length four between two vertices of f avoids $f \notin E(H)$, and f closes it. The third vertex of f may be a core of the path. □

Lemma 7. *Let H be saturated on $n \geq 3$ vertices. Then H has at most one isolated vertex, and if it has one, every other vertex has degree at least two. Consequently $3|E(H)| \geq n - 1$, and $3|E(H)| \geq 2(n - 1)$ if H has an isolated vertex. For $n \geq 4$, $|E(H)| \geq 4$.*

Proof. If z, z' are isolated, a triple $\{z, z', w\}$ contains no good pair. Let z be isolated and let v have the single hyperedge $e = \{v, a, b\}$. A Berge path of length four from a to b cannot use e (as an internal hyperedge it would need a second core of e other than a, b, namely v, which has degree one; as an end hyperedge it would join a and b directly), so with e it would form a Berge-C_5. Hence $\{z, a, b\}$ is absent with no good pair. If $|E(H)| \leq 3$ and $n \geq 4$, some triple is absent, and $H + f$ has at most four hyperedges, too few for a Berge-C_5. □

Lemma 8 (Loose expansions)**.** *Let G be a bipartite graph and H its loose expansion, with $E_{uv} = \{u, v, \ell_{uv}\}$. Then H is free, and:*

1. *two vertices of G are good in H if and only if they are joined by a path of length four in G; in particular vertices in different parts are never good;*
2. *ℓ_e and a vertex w of G are good if and only if a path of length three in $G - e$ joins an end of e to w and avoids the other end of e or ends there (so that the five cores are distinct);*
3. *ℓ_e and ℓ_f are good if and only if a path of length two in $G - e - f$ joins an end of e to an end of f, with five distinct cores in total; if $e = ux$ and $f = uy$ share u, this holds exactly when x and y have a common neighbour other than u.*

Proof. A private vertex has degree one, so it is never an internal core of a Berge path or a core of a Berge cycle; a hyperedge E_{uv} used between two vertices of G supplies the pair uv. A Berge-C_5 would therefore be a cycle of length five in G, which is bipartite. The three translations follow in the same way: the private end vertices use their own hyperedges, and the remaining part of the path consists of graph edges. For $e = ux$, $f = uy$ the middle path of length two must join x to y: using u at both ends repeats a core, and a path of length two from u to x or y has the wrong parity. □

Lemma 9 (Cloning a universal leaf)**.** *Let H be saturated and let x be a vertex of degree one, in the hyperedge $E = \{u, v, x\}$, that is good with every other vertex. Then $H' = H + \{u, v, x'\}$ with a new vertex x' is saturated. Moreover any other vertex that is good with every vertex of H is good with every vertex of H'.*

Proof. There is no Berge path of length four from u to v in H: as in the proof of Lemma 7 it could not use E, and with E it would close a Berge-C_5. A Berge-C_5 of H' uses $E' = \{u, v, x'\}$ and x' is not a core, so it would contain such a path; hence H' is free. Replacing the first hyperedge E and first core x of a Berge path of length four from x to w by E' and x' shows that x' is good with every $w \neq x$ of H. The swap of x and x' is an automorphism of H' fixing every other vertex, so a vertex good with x is good with x'. An absent triple of H' containing x' contains an old vertex other than x, which is good with x'; other absent triples keep their good pairs. □

Lemma 10 (Rooted gluing)**.** *Let $H_1, \dots, H_k$ be saturated hypergraphs that share a vertex r and are otherwise vertex-disjoint, and suppose that in each H_i every vertex other than r has a Berge path of length two to r. Then $H_1 \cup \cdots \cup H_k$ is saturated and has the same property.*

Proof. A Berge cycle cannot pass through two pieces, since it would have to traverse the cut vertex r twice; so the union is free. A Berge path between two vertices of one piece stays in that piece, so an absent triple inside a piece keeps its good pair. Any other absent triple contains non-root vertices u, w of different pieces, and their paths of length two to r form a Berge path of length four. □

3 The rooted interval family

For $p, q \geq 1$ and $t \geq 0$ let $B(p, q, t)$ have vertices a, b, c, d, e, $x_1, \dots, x_p$, $y_1, \dots, y_q$ and u_i, v_i, w_i $(1 \leq i \leq t)$, and hyperedges

$$A = abc, \quad B = ade, \quad X_j = bdx_j, \quad Y_j = cey_j, \quad U_i = bu_iv_i, \quad W_i = ev_iw_i.$$

It has $n = 5 + p + q + 3t$ vertices and $m = 2 + p + q + 2t = n - 3 - t$ hyperedges, and $B(1, 1, 0)$ is the seven-vertex block of the introduction. The map $b \leftrightarrow e$, $c \leftrightarrow d$, $x \leftrightarrow y$, $U_i \leftrightarrow W_i$, $u_i \leftrightarrow w_i$ exchanges the roles of the two sides; we call it the symmetry.

Proposition 11. *$B(p, q, t)$ is saturated, and every vertex other than a has a Berge path of length two to a. Consequently every integer m with $\lceil 2(n-1)/3 \rceil \leq m \leq n - 3$ lies in* Spec(n) *for $n \geq 7$.*

Proof. No path of length three joins b and e. No hyperedge contains both. In such a path with hyperedges E_1, E_2, E_3 and internal cores z, z' consider E_2: if $E_2 = A$ then $\{z, z'\} = \{a, c\}$ and the only hyperedge containing b and a or c is A; if $E_2 = B$ the same holds at e; every other hyperedge has a vertex of degree one among the two vertices other than b, e, which cannot be internal.

Freeness. A Berge-C_5 avoiding b as a core cannot use X_j or U_i (their only pairs avoiding b contain a vertex of degree one), then not W_i either, and d lies only in B among the remaining hyperedges; its cores would lie in $\{a, c, e\}$. By the symmetry it also uses e. The cycle then splits into two paths from b to e of lengths summing to five, neither of length one, hence of lengths two and three, which is impossible.

Saturation. Every triple contains two vertices other than b, e, or contains b and e. The following Berge paths of length four, listed by cores and hyperedges, cover every pair of base vertices other than $\{b, e\}$ together with $\{b, e\}$, up to the symmetry $(X = X_j,\ Y = Y_k)$:

pair	cores	hyperedges	pair	cores	hyperedges
a, b	a, c, e, d, b	A, Y, B, X	x_j, b	x_j, d, e, c, b	X_j, B, Y, A
a, c	a, b, d, e, c	A, X, B, Y	x_j, c	x_j, b, a, e, c	X_j, A, B, Y
a, d	a, e, c, b, d	B, Y, A, X	x_j, d	x_j, b, c, e, d	X_j, A, Y, B
a, e	a, d, b, c, e	B, X, A, Y	x_j, e	x_j, d, a, c, e	X_j, B, A, Y
b, e	b, d, a, c, e	X, B, A, Y	x_j, y_k	x_j, b, a, e, y_k	X_j, A, B, Y_k
c, d	c, e, a, b, d	Y, B, A, X	x_j, x_k	x_j, b, a, d, x_k	X_j, A, B, X_k
x_j, a	x_j, d, e, c, a	X_j, B, Y, A			

From b there are Berge paths of length three inside the base to a $(b, c, e, a$ via $A, Y, B)$, c $(b, d, e, c$ via $X, B, Y)$, d $(b, c, e, d$ via $A, Y, B)$, x_j $(b, a, d, x_j$ via $A, B, X_j)$ and y_j $(b, a, e, y_j$ via $A, B, Y_j)$, and symmetrically from e. Prepending U_i (from u_i or v_i to b) or W_i (from w_i to e) joins every extension vertex to every base vertex other than b, e. Pairs of extension vertices are joined by

pair	cores	hyperedges	pair	cores	hyperedges
u_i, u_j	u_i, v_i, e, v_j, u_j	U_i, W_i, W_j, U_j	u_i, v_j	u_i, b, a, e, v_j	U_i, A, B, W_j
w_i, w_j	w_i, v_i, b, v_j, w_j	W_i, U_i, U_j, W_j	w_i, v_j	w_i, e, a, b, v_j	W_i, B, A, U_j
u_i, w_j	u_i, b, a, e, w_j	U_i, A, B, W_j	v_i, v_j	v_i, b, a, e, v_j	U_i, A, B, W_j

with $i \neq j$ in the rows with equal letters. Lemma 6 gives saturation.

Root. Paths of length two to a: a, B, d, X_j, b and a, A, b, X_j, d; symmetrically for c, e; a, A, b, X_j, x_j; a, B, e, Y_j, y_j; a, A, b, U_i, u_i and v_i; a, B, e, W_i, w_i.

Sizes. For $0 \leq t \leq \lfloor (n-7)/3 \rfloor$ take $q = 1$ and $p = n - 6 - 3t \geq 1$; then $m = n - 3 - t$ runs over $[\lceil 2(n-1)/3 \rceil, n-3]$. □

By Lemma 10, bouquets of copies of $B(1,1,0)$ glued at a also realize $\lceil 2(n-1)/3 \rceil$; this gluing is the variant formalized in Lean (Section 8).

4 Loose expansions of bipartite graphs

Throughout, Lemma 8 translates goodness into paths in G, and Lemma 6 reduces saturation to a good pair in every absent triple. In each displayed path the cores are distinct and the graph edges, hence the hyperedges, are distinct.

4.1 Proof of Theorem 4

Write $U = V(F)$, and for $x \in X_{ij}$ call ℓ_{ix}, ℓ_{jx} *normal leaves* and $s_i = \ell_{zi}$ *spoke leaves*. Every vertex of F has two neighbours in F, and $\lambda_{ij} \geq 2$.

Vertices of G. Distinct hubs i, j: choose $k \in N_F(i) \setminus \{j\}$ and $x \in X_{ik}$; use i, x, k, z, j. The apex and $x \in X_{ij}$: choose $k \in N_F(i) \setminus \{j\}$, $y \in X_{ik}$; use z, k, y, i, x. Distinct x, y in the second part: choose ends i of x and $j \neq i$ of y; use x, i, z, j, y.

Normal leaves and vertices of G. Fix $x \in X_{ij}$. Then ℓ_{ix} is good with every hub $k \neq j$ via ℓ_{ix}, x, j, z, k (also for $k = i$), with z via ℓ_{ix}, i, y, j, z for $y \in X_{ij} \setminus \{x\}$, and with every y in the second part via ℓ_{ix}, i, z, k, y for an end $k \neq i$ of y ($k = j$ if $y = x$). So ℓ_{ix} has at most one bad vertex of G, namely j.

Spoke leaves and vertices of G. s_i is good with every hub j via s_i, z, k, x, j ($k \in N_F(j) \setminus \{i\}$, $x \in X_{jk}$), with z via s_i, i, x, j, z ($j \in N_F(i)$, $x \in X_{ij}$), and with $x \in X_{ab}$ whenever $N_F(i) \cap \{a, b\} \neq \emptyset$: for j in the intersection use s_i, i, y, j, x with $y \in X_{ij} \setminus \{x\}$.

Pairs of leaves. Two spoke leaves s_i, s_j with $ij \in E(F)$ are good via s_i, i, x, j, s_j, $x \in X_{ij}$. Normal leaves ℓ_{ix}, ℓ_{jy} with $i \neq j$ are good via $\ell_{ix}, i, z, j, \ell_{jy}$; normal leaves ℓ_{ix}, ℓ_{iy} with $x, y \in X_{ij}$ via $\ell_{ix}, x, j, y, \ell_{iy}$. The remaining normal pairs ℓ_{ix}, ℓ_{iy}, $x \in X_{ij}$, $y \in X_{ik}$, $j \neq k$, form for each i a complete bipartite graph, so the bad pairs among normal leaves form a triangle-free graph. A spoke leaf s_i and a normal leaf ℓ_{jx}, $x \in X_{jk}$, are good via s_i, z, k, x, ℓ_{jx} if $k \neq i$ and via s_i, i, y, j, ℓ_{jx} with $y \in X_{ij} \setminus \{x\}$ if $k = i$.

Every triple has a good pair. Three vertices of G contain two in one part. Two vertices of G and a normal leaf: the leaf has one bad vertex of G. Two vertices of G and a spoke leaf: if both are bad with the leaf, both lie in the second part. One vertex w of G and two normal leaves: if both leaves are bad with w, both have w as their opposite hub, so either their hubs differ or they share hub and opposite hub, and they are good. A spoke and a normal leaf are always good. One vertex w and two spoke leaves s_i, s_j with $ij \notin E(F)$: w is good with both if it is z or a hub; if $w \in X_{ab}$ were bad with both, then $N_F(i)$ and $N_F(j)$ miss $\{a, b\}$, so $a \notin \{i, j\}$ and i, j, a are pairwise non-adjacent, contradicting $\alpha(F) \leq 2$. Three leaves contain a spoke-normal pair, a good pair of normal leaves (triangle-free bad graph), or two spoke leaves adjacent in F ($\alpha(F) \leq 2$).

Counts and supports. G has $1 + h + S$ vertices and $h + 2S$ edges, giving the counts. A finite 2-regular graph is a disjoint union of cycles; a cycle of length at least six or two components other than two triangles have three independent vertices. □

4.2 Proof of Theorem 3

Necessity. Let the expansion be saturated. For distinct $u, v \in A$ and any $x \in B$, the triple $\{u, v, x\}$ is absent (each hyperedge has a private vertex) and its pairs with x have the wrong

parity, so u, v are joined by a path u, x, w, y, v of length four in G. Let J be the graph on A in which uv is an edge when some vertex of B has neighbourhood $\{u, v\}$. If u had three neighbours v_1, v_2, v_3 in J, with $N(x_i) = \{u, v_i\}$, the private vertices of ux_1, ux_2, ux_3 would form an absent triple without a good pair (Lemma 8(3)). So J has maximum degree two, and by the paths above every two vertices of A have a common neighbour in J. Hence J is a triangle and A is its vertex set. If $B_{ab} = \{x\}$, the triple $\{\ell_{ax}, b, x\}$ has no good pair: b, x have different parities, a path of length three from a to b has the wrong parity, one from x to b avoiding ax starts with xb and meets b too early, and a path of length three from a to x avoiding ax ends with bx and needs a second common neighbour of a, b. If $B_{ab} = \{x, y\}$, then x, y are not good (a path of length four between them passes through a third vertex of B_{ab}), and $\{x, y, a\}$ has no good pair.

Sufficiency. Let i, j, k denote the three hubs in some order. Hubs i, j: i, x, k, y, j with $x \in B_{ik}$, $y \in B_{jk}$. Vertices of B: x, i, z, j, y for $x, y \in B_{ij}$ with $z \in B_{ij} \setminus \{x, y\}$, and x, j, z, k, y for $x \in B_{ij}$, $y \in B_{ik}$, $z \in B_{jk}$. The private vertex ℓ_{ix}, $x \in B_{ij}$, is good with every vertex of G except j, via the following paths of length three after its own hyperedge: x, j, z, i ($z \in B_{ij} \setminus \{x\}$) to i; x, j, z, k ($z \in B_{jk}$) to k; i, z, j, y ($z \in B_{ij} \setminus \{x, y\}$) to $y \in B_{ij}$; i, z, k, y ($z \in B_{ik} \setminus \{y\}$) to $y \in B_{ik}$; i, z, k, y ($z \in B_{ik}$) to $y \in B_{jk}$. Private vertices ℓ_{ix}, ℓ_{sy} with $i \neq s$ are good via $\ell_{ix}, i, z, s, \ell_{sy}$, $z \in B_{is} \setminus \{x, y\}$; with $i = s$ and x, y in the same class B_{ij} via $\ell_{ix}, x, j, y, \ell_{iy}$; the remaining pairs form, for each hub, a complete bipartite graph. As in the previous proof, every triple contains a good pair.

Counts and isomorphism. The counts are immediate. Hubs have degree at least six, B-vertices degree two and private vertices degree one, so an isomorphism permutes the hubs and the classes; conversely all vertices of a class are interchangeable. □

4.3 Two further families

Proposition 12. *For every $n \geq 14$, $\lceil (2n-4)/3 \rceil \in \mathrm{Spec}(n)$.*

Proof. For $t \geq 3$ let G_t have parts $\{a, b, c\}$ and $\{y_1, \ldots, y_t\}$, with a, b adjacent to every y_i and c adjacent to y_1, y_2; its expansion H_t has $n = 3t + 5$ and $m = 2t + 2$. Same-part pairs: a, y_1, c, y_2, b; a, y_3, b, y_1, c; b, y_3, a, y_1, c; y_i, a, y_h, b, y_j with $h \notin \{i, j\}$. Pairs of private vertices: for ay_i, ay_j use the middle path y_i, b, y_j; for ay_i, by_j use a, y_h, b with $h \notin \{i, j\}$; for ay_i, cy_j use y_i, b, y_j if $i \neq j$ and $a, y_{j'}, c$ with $\{j, j'\} = \{1, 2\}$ if $i = j$; for cy_1, cy_2 use y_1, a, y_2; the cases with b are symmetric. So all private pairs are good. A private vertex on ay_i (or by_i) is good with every y_j via a, y_h, b, y_j; one on cy_i is good with c via y_i, a, y_j, c and with a (and symmetrically b) via y_i, b, y_h, a, $h \neq i$. Hence every triple has a good pair: three vertices of G_t contain two in one part, two private vertices are good, and a private vertex with two vertices of G_t in different parts is good with the vertex in $\{y_j\}$ (for a hyperedge at a or b) or with the hub (for a hyperedge at c). So H_t is saturated. The private vertex on ay_1 is good with every vertex: the paths above, and y_1, b, y_2, a; y_1, c, y_2, b; y_1, b, y_2, c to the hubs. The same holds for by_1. Cloning these leaves (Lemma 9) zero, one or two times gives $n = 3t + 5 + r$ and $m = 2t + 2 + r = \lceil (2n-4)/3 \rceil$ for $r \in \{0, 1, 2\}$. □

Proposition 13. *For every $n \geq 30$, $\lceil (2n-6)/3 \rceil \in \mathrm{Spec}(n)$.*

Proof. Theorem 3 with class sizes $3, 3, t-6$ gives every $n = 3+3t \geq 30$ with $(2n-6)/3$ hyperedges. Add the graph edge cz_0 for some $z_0 \in B_{ab}$, expanded by a new private vertex p. The graph stays bipartite, and p is good with every old vertex: with a hub u via z_0, v, q, u (for $u \in \{a, b\}$ take v the other of a, b; for $u = c$ take $v = a$; $q \in B_{vu} \setminus \{z_0\}$); with $z \in B$ via c, q, u, z for $u \in N(z) \setminus \{c\}$ and $q \in B_{cu} \setminus \{z\}$; with $\ell_{u,z}$, $u \neq c$, via the middle path c, q, u, $q \in B_{cu} \setminus \{z\}$; with $\ell_{c,z}$, $z \in B_{cv}$, via z_0, v, z. This gives $n = 4 + 3t$, $m = 1 + 2t$, and cloning p (Lemma 9) gives $n = 5 + 3t$, $m = 2 + 2t$; both equal $\lceil (2n-6)/3 \rceil$. □

5 Proof of Theorem 1

Proposition 11 gives $[\lceil 2(n-1)/3\rceil, n-3]$ for $n \geq 7$, and Theorem 2 gives $n-2$ for $n \geq 7$ (a cone over $C_4 \cup K_2$ at $n = 7$ and over an admissible tree with four-cycles afterwards) and $n-1$ when $4 \mid n-1$ (cones over four-cycles). Propositions 12 and 13 give $\lceil (2n-4)/3\rceil$ for $n \geq 14$ and $\lceil (2n-6)/3\rceil$ for $n \geq 30$, and Theorem 4 with $F = C_3$ gives $(2n-5)/3 = \lceil (2n-6)/3\rceil$ at $n = 25, 28$. With $F = C_4, C_5, 2C_3$ it gives $\lceil (2n-8)/3\rceil$ for $n \equiv 0 \pmod 3$, $n \geq 33$, for $n \equiv 2$, $n \geq 41$, and for $n \equiv 1$, $n \geq 49$; together these cover $n \geq 47$ and $n = 41, 44$, while for $n \equiv 0$ the value coincides with $\lceil (2n-6)/3\rceil$. Consecutive endpoints in the list differ by zero or one, since the arguments of the ceilings differ by $2/3$, so no integer is skipped. □

6 Cones

Let H be a cone with root r and link G; each hyperedge is $\{r, x, y\}$ with $xy \in E(G)$, and every pair of non-root vertices lies in at most one hyperedge.

Freeness. A Berge-C_5 without the core r is a five-cycle of G. One with cores r, a, b, c, d uses the link edges ab, bc, cd and two further distinct link edges at a and at d. Hence H is free if and only if G has no five-cycle and no path a, b, c, d with distinct further edges at a and d. A path on six vertices, a cycle of length at least six, a four-cycle with a further edge at a cycle vertex, and a triangle with a vertex of degree at least two outside it all give such a configuration. So the components of a free link are trees of diameter at most four, triangles with pendant leaves at their vertices, and isolated four-cycles; conversely these are free (in a tree the two further edges would extend the path to six vertices; a triangle with leaves has only four vertices of degree at least two in its cone; a four-cycle gives only four hyperedges; and cycles cannot cross r).

Locality. A Berge path between two vertices of the cone H_D over a component D stays in H_D, so each H_D must be saturated by itself. This fails for a triangle with leaves (the triangle's vertex triple has no good pair: there are only four vertices of degree at least two), for a star with at least two leaves (no path of length four at all), for a tree of diameter three with adjacent centres u, v and a leaf x at v (the triple $\{r, u, x\}$; a path of length four needs three internal vertices of degree at least two, but only r, u, v have that degree), and for a tree of diameter four with only two non-leaf neighbours u_1, u_2 of the centre (the triple $\{r, u_1, u_2\}$). An isolated vertex or K_2 gives a complete hypergraph. A four-cycle $abcd$ is saturated: up to symmetry r, b, c, d, a; a, r, d, c, b; a, b, r, d, c are paths of length four, with hyperedges given by the link edges ab, bc, cd, da; da, ab, bc, cd; ab, bc, da, cd. For an admissible tree with centre c, branch vertices u_i with leaves $x_{i\alpha}$ and direct leaves w_β, write $A_i = \{r, c, u_i\}$, $B_{i\alpha} = \{r, u_i, x_{i\alpha}\}$, $D_\beta = \{r, c, w_\beta\}$. The following paths (with i, j, k distinct where needed) show that every triple other than $\{r, c, z\}$ contains a good pair outside $\{r, c\}$, and that $x_{i\alpha}$ is good with r and with c; the triples $\{r, c, u_i\}$ and $\{r, c, w_\beta\}$ are hyperedges.

pair	cores	hyperedges	pair	cores	hyperedges
u_i, u_j	u_i, c, u_k, r, u_j	A_i, A_k, B_k, A_j	w_α, u_i	w_α, c, u_j, r, u_i	D_α, A_j, B_j, A_i
$x_{i\alpha}, x_{j\beta}$	$x_{i\alpha}, u_i, c, r, x_{j\beta}$	$B_{i\alpha}, A_i, A_k, B_{j\beta}$	$x_{i\alpha}, u_i$	$x_{i\alpha}, r, u_j, c, u_i$	$B_{i\alpha}, B_j, A_j, A_i$
w_α, w_β	$w_\alpha, c, u_i, r, w_\beta$	$D_\alpha, A_i, B_i, D_\beta$	$x_{i\alpha}, u_j$	$x_{i\alpha}, u_i, c, r, u_j$	$B_{i\alpha}, A_i, A_k, B_j$
$w_\alpha, x_{i\beta}$	$w_\alpha, c, u_j, r, x_{i\beta}$	$D_\alpha, A_j, B_j, B_{i\beta}$	$x_{i\alpha}, r$	$x_{i\alpha}, u_i, c, u_j, r$	$B_{i\alpha}, A_i, A_j, B_j$
			$x_{i\alpha}, c$	$x_{i\alpha}, u_i, r, u_j, c$	$B_{i\alpha}, A_i, B_j, A_j$

In the row for $x_{i\alpha}, x_{j\beta}$ we take $k \neq i$, and $i = j$ is allowed there with distinct leaves. Every non-root vertex of a four-cycle or an admissible tree has a Berge path of length two to r; in an admissible tree there is no path of length three or four from r to c, because its internal cores would be branch vertices u_i, no two of which share a hyperedge.

Compatibility. By Lemma 10, any union of four-cycles and admissible trees is saturated. Two isolated link vertices z, z' give the triple $\{r, z, z'\}$ without a good pair; an isolated vertex z and

a K_2 component xy give $\{r, z, x\}$; z and an admissible tree give $\{r, z, c\}$; two K_2 components give $\{r, x, y\}$ with x, y from different components; a K_2 component xy and an admissible tree give $\{x, y, c\}$. A K_2 or an isolated vertex together with four-cycles is saturated: the vertices of K_2 reach every four-cycle vertex by their hyperedge to r followed by a path of length three, and the only triple without a good pair among the four-cycle vertices and r is the hyperedge $\{r, x, y\}$. For $n \geq 5$ combinations of isolated vertices and K_2's only are excluded. This proves the classification.

Sizes. With $N = n - 1$, a four-cycle has four vertices and four edges, an admissible tree $v \geq 7$ vertices and $v - 1$ edges (every $v \geq 7$ occurs). With $T \geq 1$ trees, $m = N - T$ for $1 \leq T \leq \lfloor N/7 \rfloor$, all realized; with $T = 0$, $m = N$ exactly when $4 \mid N$; the other two cases give $m = N - 1$. For $N = 4, 5, 6$ the only possibilities are C_4, $C_4 \cup K_1$ and $C_4 \cup K_2$. □

7 Exact small values

For $n \leq 2$ the empty hypergraph, for $n = 3$ the single triple and for $n = 4$ the complete hypergraph $K_4^{(3)}$ (with no absent triple) are saturated; $s(3) \geq 1$ because adding the only triple to the empty hypergraph creates no Berge-C_5, and Lemma 7 gives $s(n) \geq 4$ for $n \geq 4$. Witnesses with four hyperedges on five and six vertices are the cone over C_4 and $\{012, 013, 245, 345\}$; the block $B(1, 1, 0)$ works for $n = 7$ and Proposition 11 gives $s(n) \leq \lceil 2(n-1)/3 \rceil$ for $8 \leq n \leq 11$.

The remaining lower bounds and the classifications follow from three certificates. By Lemma 7, a saturated hypergraph with m hyperedges has at most one isolated vertex, at most $3m$ vertices if it has none, and at most $\lfloor 3m/2 \rfloor + 1$ vertices if it has one. Hence it suffices to treat $m = 4$ with $n \leq 13$ and $m = 5$ with $n \leq 16$ (at most one isolated vertex), and $m = 6$ with $11 \leq n \leq 18$ (no isolated vertex). Each hypergraph without isolated vertices is encoded, after labelling its hyperedges, by the multiset of incidence masks of its vertices: masks with at least two bits are listed in nondecreasing order and the remaining incidences are filled by degree-one vertices, so every hypergraph appears. The certificates contain, for every profile in the enumeration order, an explicit Berge-C_5, an absent triple whose addition leaves the hypergraph free, or the index of a listed isomorphism class. The verifier regenerates the enumeration, rejects missing or surplus records, checks freeness of $H + f$ over all five-sets of hyperedges in all cyclic orders and all injective choices of cores, checks class membership under all permutations of the hyperedge labels, and checks that each listed class is canonical, distinct from the others, saturated and used. The record counts are 1332 (13 classes), 56688 (32 classes) and 311635 (no survivor). These give $s(8) \geq 5$, $s(9), s(10) \geq 6$, $s(11) \geq 7$ and the classifications of Theorem 5.

8 Formal verification

The Lean 4 project (Lean 4.35.0, Mathlib `56a005855ea17e0b0958f73ec801ae140b0a1679`) defines a hypergraph on a vertex type V as a finite set of finite subsets of V. A Berge-C_5 is given by an injective map $v : \mathrm{Fin}\, 5 \to V$ and an injective map e to hyperedges of H with $v_i, v_{i+1} \in e_i$; `IsBergeC5Saturated` requires three-uniformity, freeness and a Berge-C_5 after inserting any absent three-element set; and `satNumber n` is the least size of a saturated hypergraph on $\mathrm{Fin}\, n$. A *rooted piece* is a free, three-uniform hypergraph in which every absent triple contains the ends of a Berge path of length four and every non-root vertex has a Berge path of length two to the root. The project proves:

- `Realizable.add`: two rooted pieces glued at their roots form a rooted piece whose orders and sizes add (Lemma 10, with the argument that a Berge-C_5 cannot cross the cut vertex);
- `exists_saturated_interval`: for every $n \geq 14$ and every m with $2(n-1) \leq 3m$ and $m \leq n - 2$ there is a saturated hypergraph on $\mathrm{Fin}\, n$ with m hyperedges;

- `exists_saturated_lower` and `satNumber_le_lower`: for every $n \geq 7$ a saturated hypergraph with $\lfloor 2n/3 \rfloor = \lceil 2(n-1)/3 \rceil$ hyperedges, so $s(n) \leq \lceil 2(n-1)/3 \rceil$;
- `exists_saturated_cone`: $n - 1 \in \mathrm{Spec}(n)$ when $n \geq 5$ and $4 \mid n - 1$;
- `satNumber_small`: $s(n) = 0, 0, 0, 1, 4, 4, 4, 4$ for $n \leq 7$.

The pieces are $K_4^{(3)}$, the cone over C_4 and the hypergraphs $B(p, q, t)$ with (p, q, t) equal to $(1, 1, 0)$, $(2, 1, 0)$, $(3, 1, 0)$, $(4, 1, 0)$, $(1, 1, 1)$, $(2, 1, 1)$ or $(3, 1, 1)$, all rooted at a; their uniformity, freeness (by the size of four hyperedges or by an exhaustive cycle search whose completeness is proved), the paths for all absent triples and the paths to the root are Boolean searches evaluated by the kernel with `decide +kernel`, each with a soundness theorem. A finite table and an induction on the order show that every pair in the range is a sum of the pieces' pairs. No `native_decide` is used, and a module-level audit shows that every declaration depends only on `propext`, `Classical.choice` and `Quot.sound`. The endpoints below $\lceil 2(n-1)/3 \rceil$ (Theorems 4, 3 and Propositions 12, 13), the cone classification, and the values for $8 \leq n \leq 11$ are not formalized; they have the conventional proofs above, an exhaustive standard-library check of every claimed pair (n, m) for $7 \leq n \leq 60$, and checked certificates.

9 Related work and scope

The definitions and the bound with coefficient $4/3$ are from [2], whose journal version is [3]; the inspected statement is that of the preprint, as the journal text could not be accessed. Gluing at a vertex is used in [2]. Linear growth is due to [4]; Gerbner, Patkós, Tuza and Vizer [6] and English, Kritschgau, Nahvi and Sprangel [5] give further linear bounds by general constructions, and Xie [8] proves $\mathrm{sat}_k(n, \text{Berge-}F) = O(n)$ for all graphs F and all k; these do not state cycle-specific coefficients below $4/3$ for this problem. Cao, Lai and Wang [1] treat Berge five-stars and K_4. Wang and Zhang [7] study the different linear saturation problem. Searches in October 2026 of arXiv, journal records and general web indexes did not find the coefficient $2/3$, the cone classification or Theorems 3 and 4; some recent full texts were inaccessible, and this does not establish priority.

Research process and AI disclosure

This is fully autonomous agentic research. Dev D. Goyal is responsible for publication. No expert validation or external peer review is claimed, and the novelty assessment is provisional.

The constructions, the cone classification, Theorems 3 and 4 and the finite certificates were found and proved by autonomous AI research agents, which also searched the literature. Their records do not name the system used; the research contract specifies an Astra referee for the internal review, which was an AI-assisted review of the proofs and not peer review, and the records state that no Lean formalization was attempted.

Opus 5.5 by Anthropic used with Claude Code assessed the package for this release. It checked the written proofs, compared the statements with [2, 4], rechecked every construction with its own exhaustive checker, re-encoded the certificates and checked them with its own verifier, wrote and ran the Lean formalization described in Section 8, and wrote this note. AI checks are not human peer review.

References

[1] X. Cao, Y. Lai and C. Wang, *Saturation in three-uniform Berge hypergraphs: the five-star spectrum and the K_4 saturation number*, arXiv:2610.32353 (2026).

[2] S. English, N. Graber, P. Kirkpatrick, A. Methuku and E. C. Sullivan, *Saturation of Berge hypergraphs*, arXiv:1710.03735v1 (2017).

[3] S. English, P. Gordon, N. Graber, A. Methuku and E. C. Sullivan, *Saturation of Berge hypergraphs*, Discrete Mathematics **342** (2019), 1738–1761. doi:10.1016/j.disc.2019.01.031.

[4] S. English, D. Gerbner, A. Methuku and M. Tait, *Linearity of saturation for Berge hypergraphs*, European Journal of Combinatorics **78** (2019), 205–213. arXiv:1807.06947.

[5] S. English, J. Kritschgau, M. Nahvi and E. Sprangel, *Saturation numbers for Berge cliques*, arXiv:2301.02973 (2023).

[6] D. Gerbner, B. Patkós, Zs. Tuza and M. Vizer, *On saturation of Berge hypergraphs*, arXiv:2103.08437 (2021).

[7] C. Wang and J. Zhang, *Linear saturation numbers of Berge-C_3 and Berge-C_4*, arXiv:2306.06435 (2023); Applied Mathematics and Computation **474** (2024).

[8] T. Xie, *A linear upper bound for Berge saturation numbers*, arXiv:2610.11409 (2026).